

30. Answer: C

Decay constant, using $A = \lambda N$,

$$\lambda = \frac{A}{N}$$

$$= \frac{3.7 \times 10^7}{4.8 \times 10^{20}}$$

$$= 7.708 \times 10^{-14} \text{ s}^{-1}$$

Using $N = N_0 \exp(-\lambda t)$,

$$1.5 \times 10^{19} = 4.8 \times 10^{20} \exp(-7.708 \times 10^{-14} t)$$

$$t = 4.5 \times 10^{13} \text{ s}$$